TABLE OF CONTENTS

1 – INTRODUCTION

1-1 Overview

This procedure will describe how to adjust the LCD display monitor to match the camera. Once the LCD display monitor is re-calibrated, it is essential to check and possibly re-calibrate the camera.

1-2 Camera Vendor Participation

It is recommended that this procedure is performed with the Camera Vendor field engineer, present, should any camera adjustment be necessary. To optimize customer satisfaction, it is also recommended that you have one of the Customer's filming specialists available for the fine tuning and quality review of the film/LCD display monitor conformance.

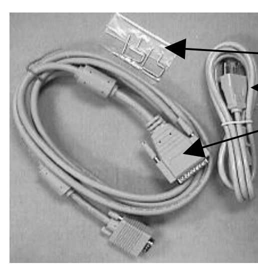
2 – NEC 1990SX MONITOR INSTALLATION

2-1 Installation of the NEC 1990SX in a Fixed Site Environment

1. Shutdown Signa and perform a safety lockout/tagout of the operator workspace. See <u>Safety</u> procedure to accomplish this.
2. If necessary, remove the old LCD or CRT from the desktop.
3. Remove the new LCD monitor from its box and its packing and set the NEC 1990SX LCD color monitor on the workspace desktop.
4. Some items will not be used. Dispose of them locally. See Illustration 1-1.

Discard:
1. DVI cable
2. Standard VGA cable
3. PC CDROM drivers

Use:
Restraint Clips
Power Cable
SVGA-RGB Cable

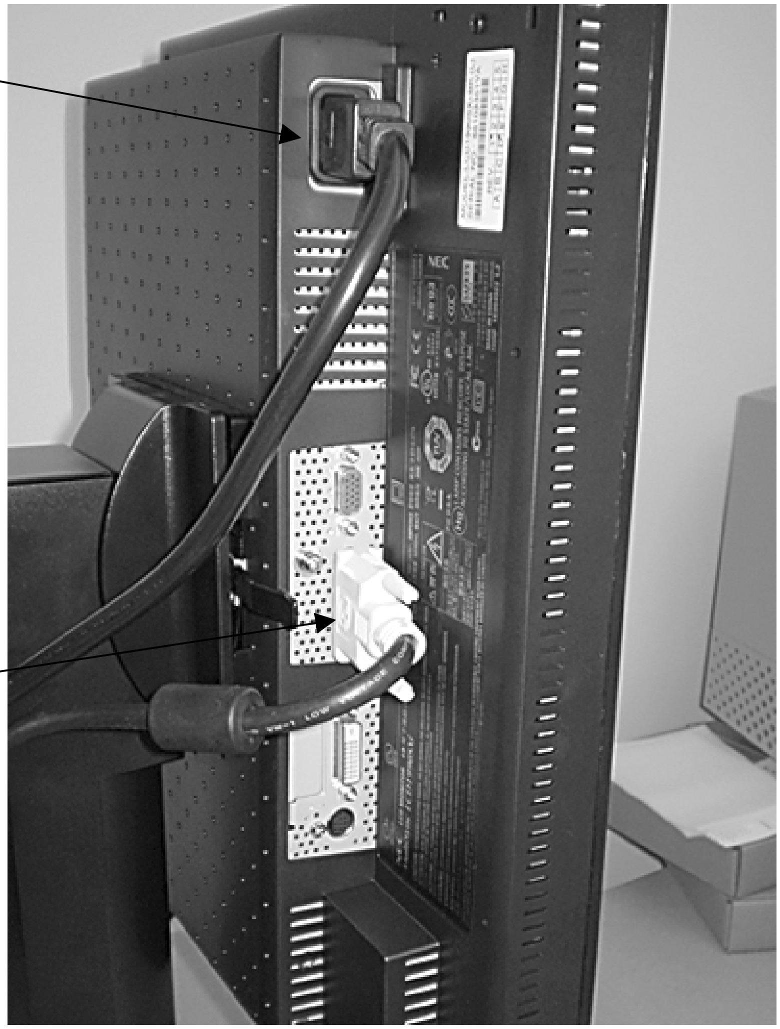


If you are replacing a Host Monitor make sure you also order: GE Part# 2376768
This CDROM disk is labeled "MR Signa® Host Monitor Gamma Correction Software" contains the most up-to date gamma files needed to setup gamma. <u>The software on the CDROM is required, to setup gamma on a replacement monitor</u>. Follow the installation instructions included with the CDROM disk.

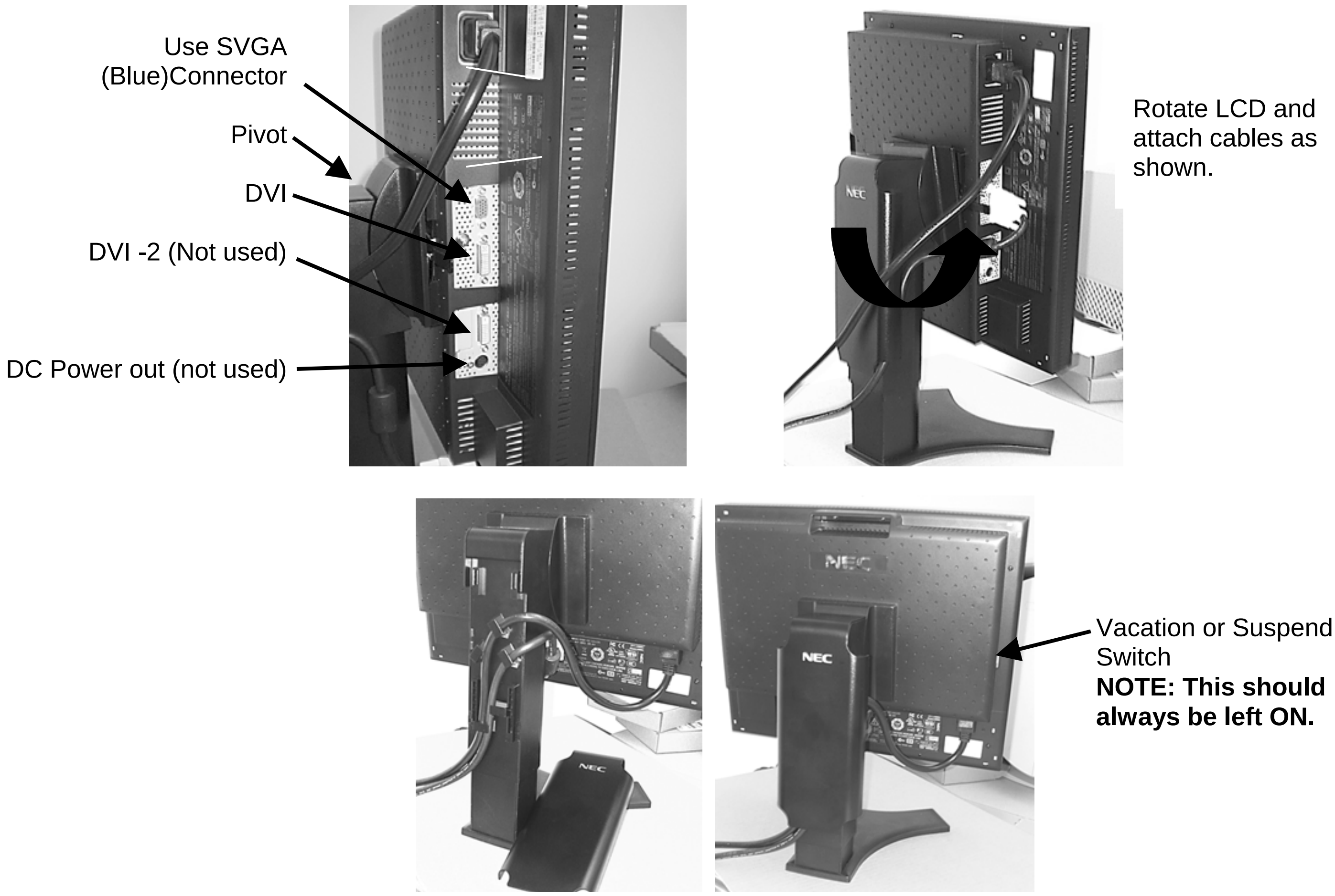
GE Gamma Patch CDROM
(GE Part# 2376768)

DISCARD EXTRA CABLES AND PC CDROM DISK
ILLUSTRATION 1-1

5. Rotate the monitor and connect the video and power cables neatly to the back of the monitor base and the operators workspace. Secure the cables with the restraint clips provided. See Illustrations 1-2 and 1-3.



VIDEO AND POWER CABLE CONNECTIONS
ILLUSTRATION 1-2



VIDEO AND POWER CONNECTED RESTRAINT CLIPS IN PLACE
ILLUSTRATION 1-3

6. Remove the Safety Lockout of the operators workspace and restore power to the system.
7. Verify that main LCD monitor power is ON. If not, press the power button on the front panel and the side panel to turn the main LCD monitor ON. See Illustration 1-3 for side view.
8. The monitor should be positioned no closer than 16 inches and no further away than 28 inches from your eyes. The optimal distance is 24 inches for all monitor types.

Note

Allow the monitor to warm-up for 20 minutes before performing any adjustments.

9. Boot up Signa.
10. Password: **`adw2.0`** `[Enter]`

2-2 Installation or Replacement of an NEC 1990SX in a Mobile Environment

This section pertains to mobile systems only. This procedure will only work for the NEC 1990SX.

1. Remove the replacement LCD panel from its shipping box.
2. Place the new LCD Monitor face down on a flat surface where it will not get scratched or damaged.
3. Remove the 4 screws that hold the new LCD panel to its base.
4. Remove the old NEC 1990SX monitor from its base in the same manner.
5. Using a Phillips screwdriver, mount the new LCD panel to the existing mobile base using the 4 screws.
6. Attach the old LCD Panel to the new base or dispose of the both items according to normal procedures.

3 – VERIFY AMBIENT LIGHTING CONDITIONS

LCD's are extremely susceptible to viewing angle. All adjustments and image comparisons should be done from exactly the same viewing angle in all directions. Sit directly in front of the monitor at all times.

In the review area and operator workspace area, verify that the ambient lighting conditions are adjusted to a minimum level. In the operator workspace area, there should be only sufficient light for safely operating the system.

In the review area and operator workspace area, verify that light-boxes are not emitting light, or are properly masked, when not displaying film. This will be a source of excessive glare.

In both review area and operator workspace area, verify that there is no source of glare for reviewing films or setting up the images for film. For example, windows should not allow direct light (blinds should be closed).

Use the same type of artificial lighting for the operator workspace area and the review area.

4 – SETTING LCD MONITOR FREQUENCY RATE AND RESOLUTION

Failing to do this section will result in image ghosting and unnecessary troubleshooting.

4-1 Setting the Line Rate and Resolution

WARNING!

PERFORM THIS PROCEDURE <u>EVERY TIME SOFTWARE IS RELOADED</u>. LINE RATE AND GAMMA SETTINGS ARE NOT CAPTURED ON THE SAVEINFO MOD AND ARE NOT CAPTURED IN THE HEALTH PAGE STATUS OF THE SYSTEM.

MONITOR *LINE RATE* OR *REFRESH RATE* IS REFERRED TO AS *FRAME RATE* OR *SWAP RATE* ON SGI SYSTEMS. FOR SIMPLICITY, THE TERM *LINE RATE* WILL BE USED IN THIS DOCUMENT.

Note

Failure to set the Line Rate to 60Hz for ALL LCD Host monitors will result In Image Ghosting and customer complaints. This procedure needs to be done EVERY TIME software has been reloaded. Line Rate is NOT captured on the SavInfo MOD or in the Health Page Status of the system.

Note

UNIX/IRIX is case sensitive. Use the commands exactly as shown as “root owner”.

1. From the desktop, click on the **C-shell tool**.
2. Switch to root owner. Type: **`su root`** `[enter]`; Enter your password.
3. Type: **`chkconfig fixed72hz off`** `[enter]`.

Note

If you get the following error, you may have incorrectly typed the command. Re-type the command. If the error appears the second time, ignore it. Proceed with the step 4.

```
chkconfig: cannot set flag "fixed72hz": No such file or directory
use "chkconfig -f fixed72hz off" to create and set the flag
```

4. Type: **`/usr/gfx/setmon –x –w 1280x1024_60`**
5. Type: **`exit`** `[enter]` to exit root
6. Type: **`exit`** `[enter]` to exit the C-shell
7. Select **System Shutdown**, (OK to shutdown) (Reboot is essential)
8. Select **Restart.**
9. Choose **Signa icon** and enter your password.
10. Proceed to Section 5 – Signa Host Gamma Setup, NEC 1990SX.

5 – SIGNA HOST GAMMA SETUP, NEC 1990SX

The gamma value is modified to optimize the contrast level of the image mid-tones to more closely represent the same contrast that is filmed.

Note

At time of System Installation:

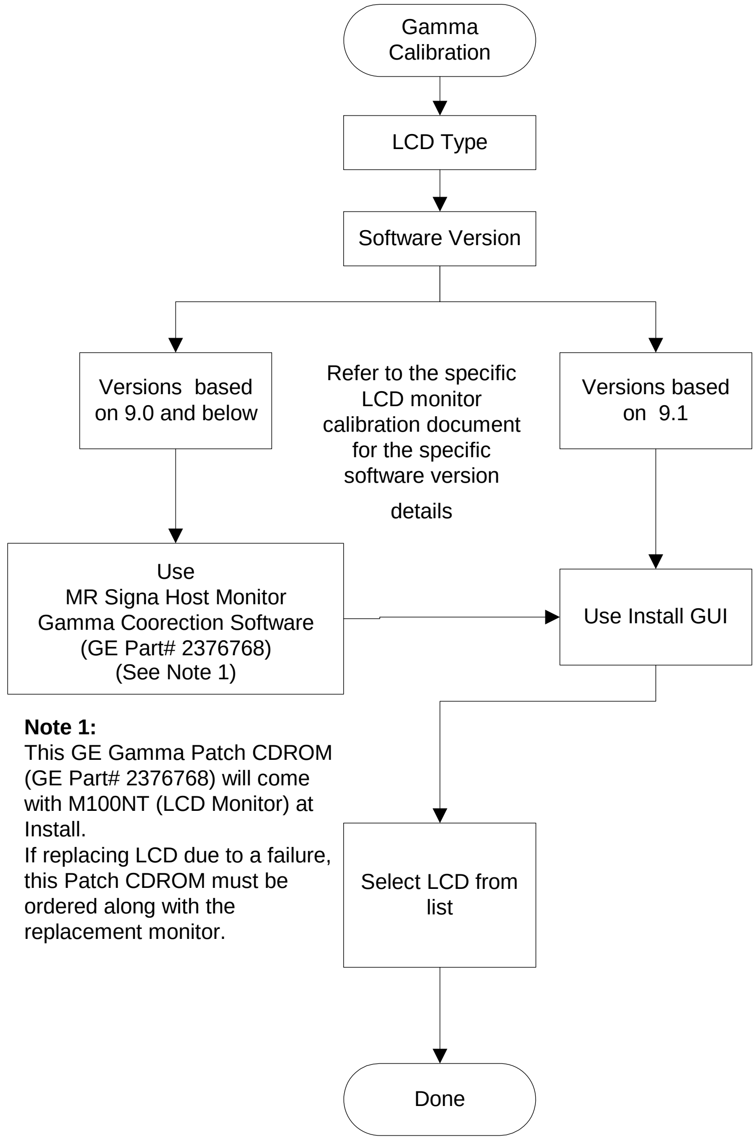
As of July 28, 2003, all NEW LCD monitors will have a Gamma Software CDROM disk INCLUDED in the box, (GE Part # 2376768) which should be used for software versions based on 9.0 and below. Use this CDROM disk and the instructions it contains to set up gamma for those software versions.

Upon replacement of a failed monitor, make sure you also order:

GE Part # 2376768. The CDROM disk labeled *MR Signa® Host Monitor Gamma Correction Software* contains the most up-to-date gamma files needed to set up gamma. *The software on the CDROM is required to install a replacement monitor*. Follow the installation instructions included with the CDROM disk.

WARNING!

MR SIGNA® HOST MONITOR GAMMA CORRECTION SOFTWARE (GE PART # 2376768) SUPERCEDES ALL OTHER GAMMA FILES THAT MAY BE ON VERSION 14 AND 15 LX/8X SERVICE METHODS CDROM.



Note 1:
This GE Gamma Patch CDROM (GE Part# 2376768) will come with M100NT (LCD Monitor) at Install.
If replacing LCD due to a failure, this Patch CDROM must be ordered along with the replacement monitor.

PROCESS FLOW CHART
ILLUSTRATION 4-1

5-1 Gamma Configuration for all SGI Computers

Select the section to use based on your computer type and software version. See Table 5-1.

GAMMA PROCEDURES BY SECTION

TABLE 5-1

Computer Type:	Software Version:	Gamma Setting:
All SGI computers	ALL Software versions 9.0 and below. Setting Gamma *with* the CDROM disk that came with the monitor.	**See Section 5-1-1**
Octane 1 and Octane 2	9.1, HFO2, HF03 Setting Gamma on monitors *without* a CDROM Disk	**See Section 5-1-2**

5-1-1 Setting Gamma with the Patch CDROM – All Software versions 9.0 and below

- If you are Installing the NEC 1990SX monitor for the first time a CDROM labeled *MR Signa® Host Monitor Gamma Correction Software* is included in the Monitor box. The installation instructions are included with the CDROM disk and should be followed to load the gamma software on the system.
- If you are replacing a failed LCD Monitor, then a CDROM labeled *MR Signa® Host Monitor Gamma Correction Software*, (GE Part # 2376768) should have been ordered along with the replacement monitor. The installation instructions are included with the CDROM disk and should be followed to load the gamma software on the system.
- This latest CDROM disk should be retained and re-used any time a reload of software occurs on the system.
- If you do not know the software version, type `getver` in a c-shell.

This option is used to configure an NEC 1990SX LCD. To set gamma:

If you are replacing the NEC 1990SX with another 1990SX and have not reloaded software: No gamma changes are necessary. Proceed to Section 6 in this document.

If you are replacing the NEC 1990SX with the another 1990SX and have reloaded software: Proceed to step 1 below:

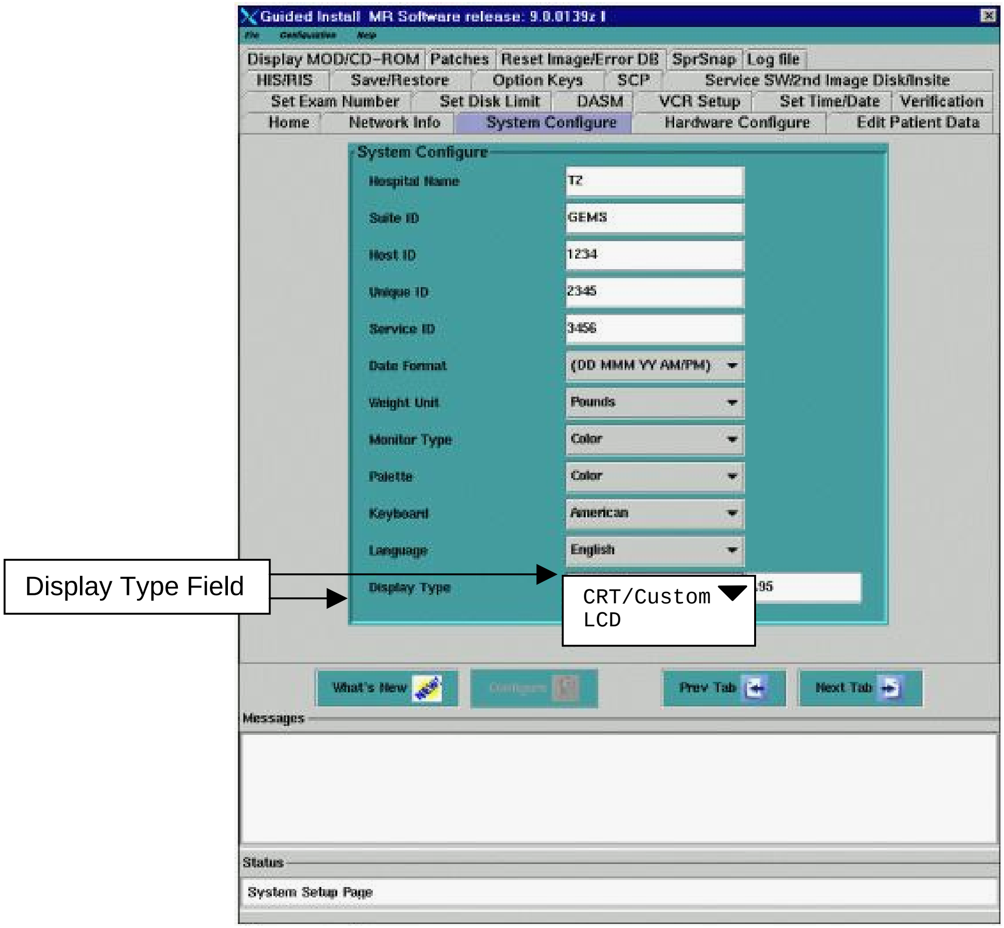
1. Insert the *MR Signa® Host Monitor Gamma Correction* CDROM into the Host CDROM/DVDROM drive.
2. Follow the instructions included with the CDROM to set the gamma for your monitor type.
3. Proceed to Section 6 in this document.

5-1-2 Setting Gamma – 9.1, HF02 and HFO3

This option is used to configure an NEC 1990SX LCD. To set gamma:

- **If you are replacing the NEC 1990SX with a 1990SX and have not reloaded software:** No gamma changes are necessary. Proceed to Section 6 in this document.
- **If you are replacing the NEC 1990SX with the another 1990SX and have reloaded software:** Proceed to Step 1 below:

1. Open Guided Install and select **System Configure** tab. See Illustration 5-1.
2. Look for the **Display Type** field. See Illustration 5-1.



GUIDED INSTALL GUI SYSTEM CONFIGURE TAB- DISPLAY TYPE FIELD
ILLUSTRATION 5-1

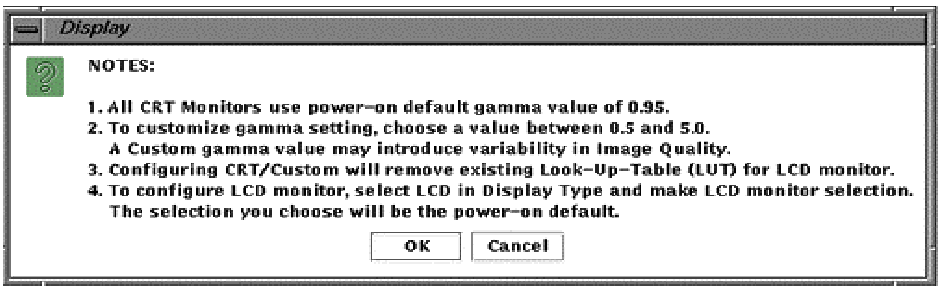
The **Display Type** field has two choices:

- **LCD** – This should be used to configure gamma for a specific LCD listed in the dropdown box (preferred method).
- **CRT/Custom** – This should be used to set a custom CRT gamma setting.

Note

It may be necessary to toggle between the **CRT/Custom** and **LCD** options in the **Display Type** field to select **LCD**.

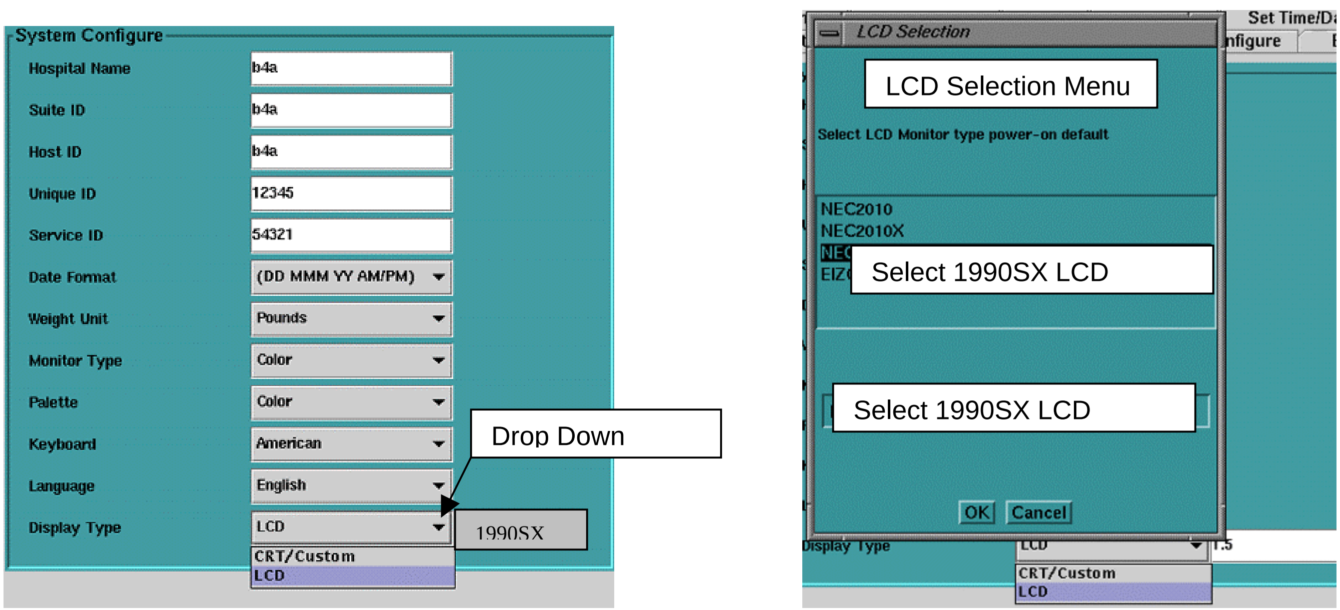
3. If **CRT/Custom** is displayed in the field:
   a. Click on the **Display Type** field.
   b. Drag your mouse over **LCD.**
   c. A popup message box will appear. Select **OK** (see illustration 5-2).



MESSAGE BOX
ILLUSTRATION 5-2

The field should change to [LCD].

   d. Select **LCD** again. The following menu will appear:



MONITOR CHOICES
ILLUSTRATION 5-3

   e. Scroll to the correct display option (1990SX) and release the mouse button. The display selected will appear in the box to the right.
   f. Click on the **Configure** button on the Install GUI to save the changes. You may see a slight change in the screen intensity when configuration is complete.
   g. A popup message box will appear. Select **OK**.
   h. Select **File** and **Quit** and **OK** from the Install/Reconfigure GUI window.

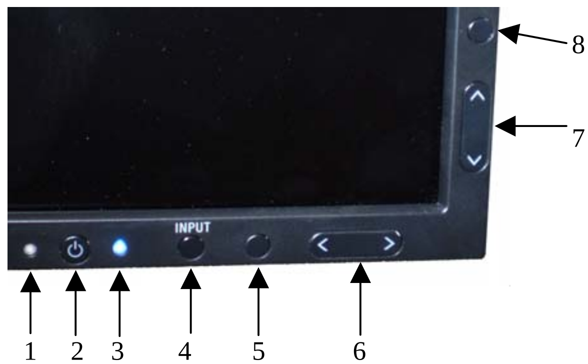
i. **Reboot the System**. A reboot is required to activate the changes. This completes the gamma calibration process.

j. Proceed to Section 6 – LCD Auto Adjustment and OSM™ On-Screen Manager Operations.

6 – LCD AUTO ADJUSTMENT AND OSM™ ON-SCREEN MANAGER OPERATIONS

This section explains:

- Basic operation of the monitor controls.
- Button panels and the On-Screen Manager setup.
- Adjusting the monitor to a default condition to proceed with calibrations.



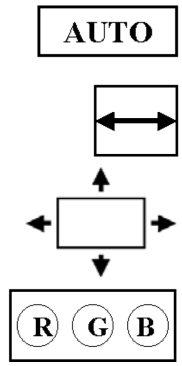
NEC 1990SX LCD FRONT PANEL OSM ACCESS BUTTONS
ILLUSTRATION 6-1

TABLE 6-1
OSM ACCESS BUTTON FUNCTIONS

1	**AMBIBRIGHT**	Detects the level of ambient lighting allowing the monitor to make adjustments to various settings, resulting in a more comfortable view experience. Do not cover this sensor.
2	**POWER**	Turns the monitor on and off.
3	**LED**	Indicates that the power is one. Can be changed between blue and green in the Advanced OSM Control menu.
4	**INPUT / SELECT**	Enters the OSM Control menu. Enters OSM submenus. Changes the input source when not in the OSM Control menu.
5	**EXIT**	Exits the OSM submenu. Exits OSM Control menu.
6	**LEFT / RIGHT**	Navigates to the left or right through the OSM Control menu.
7	**UP / DOWN**	Navigates up or down through the OSM Control menu.
8	**RESET / ROTATE OSM**	Resets the OSM back to factory settings. Pressing when OSM is not showing rotates the OSM Control menu between portrait and landscape mode.
9	**KEY GUIDE**	The Key Guide appears on the screen when the OSM control menu is accessed. The Key Guide will rotate when the OSM Control menu is rotated.

6-1 Overview

Perform the following adjustments:



1. Auto Adjust

   a. Image Adjust (automatic)

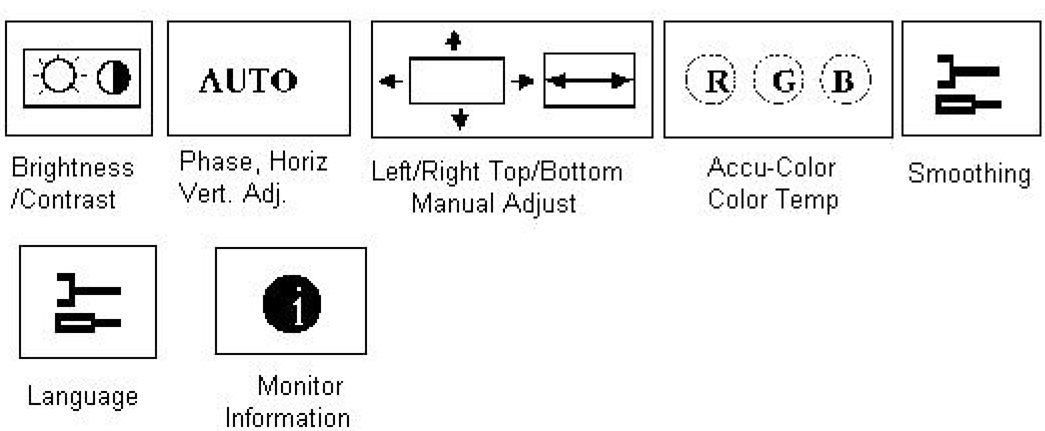
   b. Position Control (automatic)

2. AccuColor™ set-up (9300k)

3. Readjust contrast and brightness to customer's recommendation.

6-2 Using the On-Screen Manager

1. To access On-Screen Manager OSM™, press one of these buttons: **EXIT**, **LEFT**, **RIGHT**, **UP**, or **DOWN**.



MONITOR ADJUSTMENT ICONS
ILLUSTRATION 6-2

Note

To rotate the OSM menu between Landscape and Portrait modes, press the **OSM** button On, then press the **Exit** button, then press **OSM** button again.

2. Select the **Select / 1--2** button on the front of the monitor.

3. Push either the left or right control buttons (**< >**) to initiate the Auto Adjust of the Phase/ Vertical and Horizontal Centering.

Note

Manual adjustment of the Position Control and Image Adjust H. Size/Fine controls is rarely needed. Automatic adjustment takes care of this.

4. Select the **Select / 1--2** button on the front of the monitor
5. Select the **AccuColor® Control System** icon R G B and use the **< >** Control buttons to select the first entry row under this icon.
6. Select: **1 (9300)** (Monitor Color Temperature).
7. Select the **Select / 1--2** button on the front of the monitor.
8. Leave **Smoothing** setting at **Default**.
9. Select the **Select / 1--2** button on the front of the monitor.
10. Leave **Language** setting at **Default**.
11. Select the **Select / 1--2** button on the front of the monitor.
12. Leave **Display mode** setting at **Default**. Select the **Select / 1--2** button on the front of the monitor.
13. Press the **Exit** button. All adjustments are now saved.

These settings will remain in effect even if power is removed, or system software is re-booted, unless manually changed by the operator in the OSM , On-Screen Manager.

The monitor adjustment is complete. The LCD Display Monitor is now ready for integration to Signa.

Note

For further information on the On-Screen Manager and button identification, refer to the NEC LCD 1990SX User's Manual that came with the LCD Display.

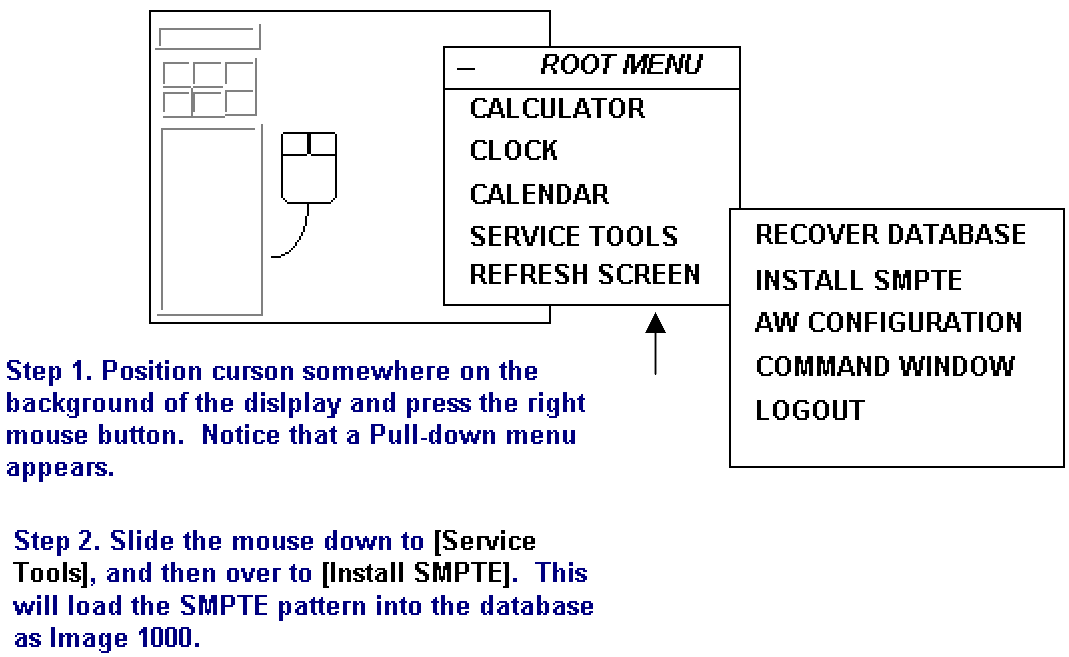
7 – FINAL ADJUSTMENT OF CONTRAST AND BRIGHTNESS

The SMPTE (Society of Motion Picture and Television Engineers) test pattern is used to provide a standard image for calibrating the Window and Level of the display for an MR scanner.

7-1 Displaying the SMPTE Pattern

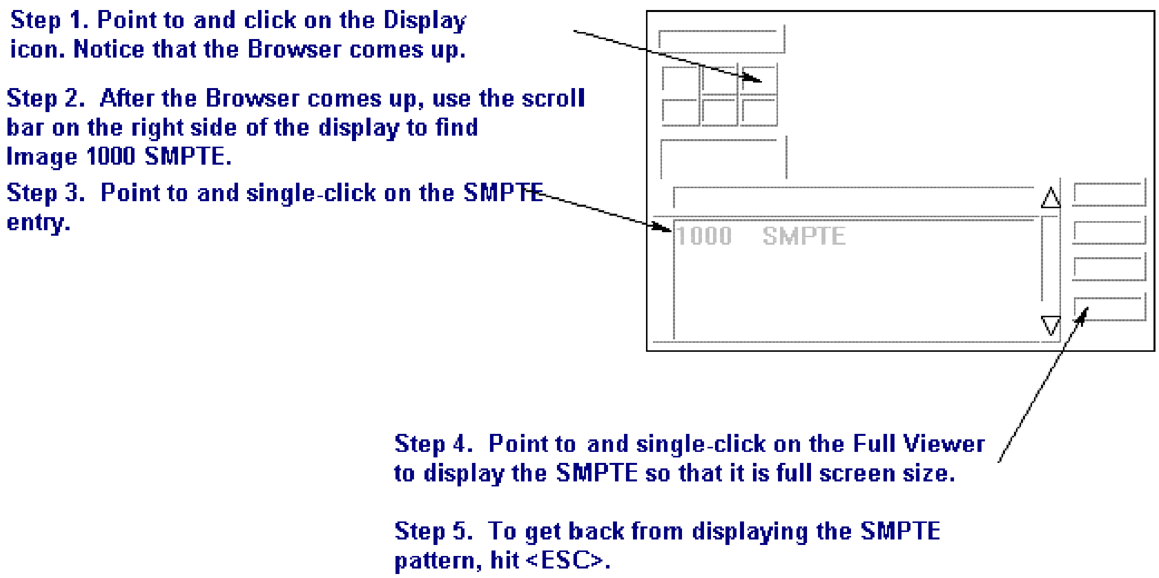
The Window and Level adjustments must now be set to ensure site-to-site uniformity in image appearance. The SMPTE test pattern is available on the Operator Workstation Host Computer after IRIX, (the operating system) has been booted, and after you have logged into the system.

1. Install and display the SMPTE pattern. Follow the steps in Illustration 7-1 below:



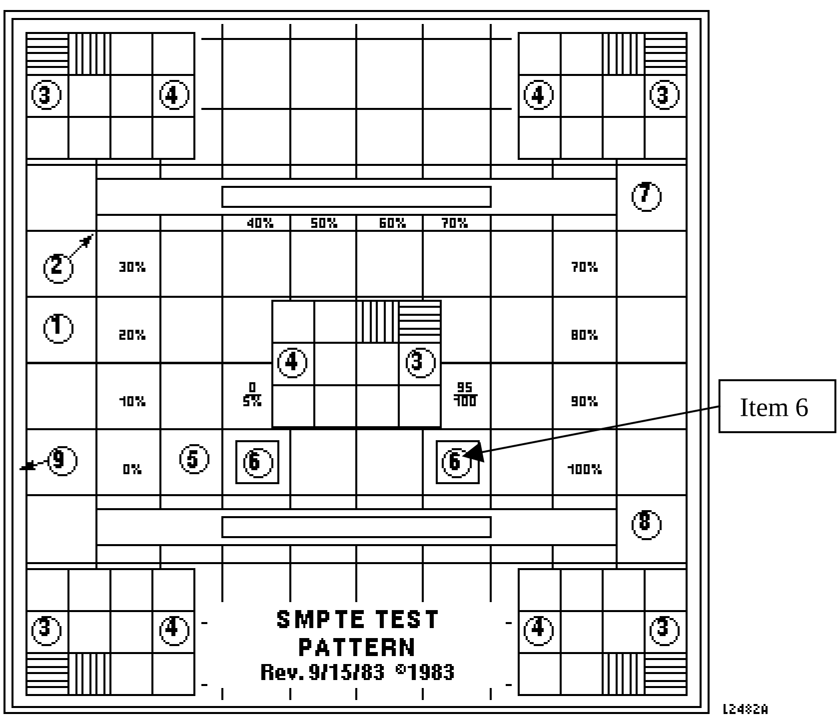
INSTALLING THE SMPTE PATTERN
ILLUSTRATION 7-1

2. Display the SMPTE Pattern by following the steps in illustration 7-2 on the next page. Switch to the browser and select the SMPTE pattern from the patient list and display it.
3. If necessary, use the **FORMAT** option to select **Full Screen** mode.



INSTALLING THE SMPTE PATTERN
ILLUSTRATION 7-2

7-2 Adjusting for Proper Window and Level Setting



SAMPLE SMPTE TEST PATTERN
ILLUSTRATION 7-3

1. With the cursor inside the displayed image, hold down the middle mouse button and move the mouse in the horizontal plane. View the window control value at the base of the image and **set window control value to 100**.
2. With the cursor inside the displayed image, hold down the middle mouse button and move the mouse in the vertical plane. View the level control value at the base of the image and **set level control value to 1024**.

7-3 Fine-Tuning the Contrast and Brightness Using the SMPTE Pattern

In this section, you will adjust the contrast and brightness controls of the LCD nonitor to optimize the SMPTE Test Pattern for the Window and Level controls.

Minimal adjustment should be necessary at this time. However, it is important that you fine-tune the Contrast and Brightness level using an industry-standard test pattern.

This is an iterative process: first, you adjust the contrast, then the brightness, and then back to the contrast.

Note

To reduce visible tearing or smearing of the pattern, LOWER the contrast.

1. To access On-Screen Manager OSM™, press one of these buttons: **EXIT**, **LEFT**, **RIGHT**, **UP**, or **DOWN**.
2. Use the **Control** buttons (**<** **>**) on the front of the monitor to adjust the brightness and contrast.
3. Look at the pattern. If the 5% and 95% patches are not visible.(item 6 in Illustration 7-3), adjust the brightness control of the display monitor until you can just barely see them. You may also need to re-adjust the contrast, if tearing or smearing of the pattern occurs. (items 1 and 2, and 5 through 8, Illustration 7-3). Continue adjusting both brightness and contrast until the image is as crisp and clear as you can make it, and the 5% and 95% patches are just barely visible.
4. When satisfied with the display, exit the menu to save the Contrast/Brightness settings.

Note

These settings will remain in effect even if power is removed or the system is re-booted. Because an operator can tamper with the brightness and contrast adjustments, make all operators aware of the affect they will have on system image quality and the camera images.

Note

If the operator chooses to deviate from this brightness and contrast adjustment, make sure the camera imaging and calibration has NOT been affected.

5. Remove the SMPTE pattern by selecting a different exam from the browser.

8 – CAMERA CALIBRATION

This procedure describes the steps necessary to verify and set the camera parameters. Once the display is re-calibrated, it is essential to re-calibrate the camera before the system is used for filming. Although a qualified GE Service Engineer could perform the steps below, it is recommended that the following procedure be performed with the on-site assistance of the camera vendor field engineer.

8-1 DASM Interpolation Setup

1. Select the **Install** soft key from the service desktop. When prompted, login using the password **operator**.
2. Select the **DASM** folder from the top of the Guided Install GUI.
3. Set the DASM Interpolation method to **linear**.
4. Before exiting the GUI, insert the SaveInfo MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **File** and then **Save GI Configuration to MOD**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for use in the next software install.
5. Exit the install GUI by selecting **File** then **Quit**. If any error conditions still exist, you will be warned that no changes will be made before exiting.

8-2 Camera Imaging Look-up Table

Verify with the camera vendor field engineer that the currently installed camera lookup table is designed to provide perceivably linear contrast for the light box conditions in the customer's viewing area. If not, request that the camera vendor field engineer replace it with the appropriate look-up table.

8-3 Camera Maximum Optical Density

Note that for optimal reviewing, the light box luminance of the diagnostic region of the film should be in the range of 50 to 500 nits. Vary the maximum optical density setting of the camera to compensate for the light box and to meet this value. A good starting position is a maximum density of 2.8.

Note

The final OD settings may be refined by the radiologists performing the image review.

8-4 Camera Contrast

1. With the maximum /minimum optical densities set to compensate for the review area's light box, select a look-up table for your camera that will produce a perceivably linear gray scale for the same light box and the overall ambient light conditions of the viewing area.

Note

The DICOM 3.14 Standard specifies the Barten's curve for linear perception. It is recommended that the manufacturer base perceptual linearity on this curve.

2. Film the SMPTE pattern on a 1-on-16-format display. Verify that the 5% and the 95% levels are visually equivalent. If not, perform a contrast test with the SMPTE pattern. Select the new contrast setting from the contrast image set. A good value for the Imation DryView is 3. Use the camera's calibration procedure to set the contrast setting. Ensure that the camera maintains a perceivably linear gray scale.

Note

Filming the SMPTE pattern for contrast calibration may be optional for the camera vendor.

3. Ask the technologist to display a clinical image and set window and level controls for the desired appearance. A good image to start with is a sagittal or axial head image.
4. Capture the image on the keypad or host control interface.
5. Print a contrast test film. Ask the technologist to select the image that best matches the displayed image on the monitor.
6. Observe the image number below the selected image and set the contrast control to this value.

8-5 Anatomical Filming

This portion of the procedure requires the technologist to verify the camera settings with true anatomical images.

1. Film representative anatomical images to confirm the settings. The image set should include T1 and T2 head images, joint images and c-spines.
2. Observe the accuracy of the low-tones, mid-tones and high-tones. If a filmed image is found to not be equivalent, then re-calibrate the camera based on the customer's evaluation.

9 – TROUBLESHOOTING GUIDE

The following section provides suggestions for troubleshooting the LCD monitor:

9-1 Explanation of Default Gamma Settings

The default gamma setting is different for different software versions. Table 9-1 indicates the suggested nominal settings. The software default settings are set at boot up. When a look-up table is installed, these default settings are ignored. If the **`setgamma`** command is used, the colorPix.cfg file and the boot script (Startmon) is updated with the new single value.

TABLE 9-1
DEFAULT GAMMA SETTINGS

LCD Monitor	Nominal Gamma Value
NEC 1990SX	0.79

As the gamma setting gets lower, the display will become dimmer (less contrast).

9-2 Complaint: Poor SNR, Dynamic Range Lack of Grayscale.

It is important to note that the software default gamma setting of 1.5 causes extremely poor image quality on both the LCD and CRT. Customers complain of ghosting or poor SNR or lack of dynamic range in the grayscale. Before troubleshooting any system problem related to these symptoms check the monitor gamma calibration, or make sure you have the correct monitor selected.

Gamma calibration tables (LUT's) are unique from one monitor type to another. DO NOT use a gamma table designed for a different monitor. If the gamma table or monitor type does not exist, start with a single gamma value of 0.95. It may be necessary to tweak this setting for your customer. Use the **`setgamma`** command to do so.

9-3 Complaint: Ghosting or Bluring images or Sheering alphanumerics LCD 's ONLY.

Ghosting and blurring of areas of gray in an image can be related to system problems, but also can be attributed to monitor Frequency Rate (Frame and Swap rates). *ALL LCD's should be set to 60Hz.*

Checking the Frame Rate and Swap Rate Setting:

1. From the desktop, click on the **C-shell button**.
2. To switch to root owner, type: **`su root`** `[enter]`; Enter your password.
3. Type: **`toolchest&`** `[enter]`.
4. Click on **Find Control Panels**.
5. Double-click on the Display icon to open **Display** tools.
6. Verify: **Active Size (Pixels): = 1280x1024 and FRAME RATE and SWAP RATE =60.00 Hz** (see Illustration 9-1).

FRAME RATE AND SWAP RATE
ILLUSTRATION 9-1

7. If it is the Frame Rate and Swap Rate are not set to 60Hz:
8. Select **1280X1024_60** from the left menu.
9. Click on the **Load** button.
10. Click on **OK** when the pop-up box asks you if you want to Load the setting now.
11. Click on **OK** when the pop-up box asks you if you want to keep the Format.
12. Click on **OK** when the pop-up box asks you if you want to make this setting the Power-On Default.
13. Close the Graphics Backend window: Click on **File**>**Exit**.
14. Close the Control Panel window: Click on **Page**>**Exit**.
15. Close Toolchest Window: Right-click and hold on the word **Toolchest** at the top of the toolchest window.
16. Drag the mouse to **Close**.
17. In the C-shell type: **`exit`** `[enter]` to exit root.
18. Type: **`exit`** `[enter]` to exit the C-shell.
19. Reboot the system.

9-4 Complaint: Camera film has changed after monitor replacement.

Note

If there is no remote camera in the list, you will not be able access the Film Composer Menu.

1. Make sure that the Monitor and Camera Calibration procedures have been performed completely and the correct Monitor Type has been selected and installed.
2. Set the camera to **Replicate** instead of linear mode, or vice versa.
   - For Kodak cameras, set to **Bi-linear** mode.
   - For other cameras try setting **Replicate** mode.
     Procedure:
     a. Open the Image Browser.
     b. Open Film Composer.
     c. Select **Configure**.
     d. Select remote printer being used by the site.
     e. Select **Update**.

f. Under magnification type, select the **Replicate** or **Bi-Linear** option. (For Kodak cameras, try **Bi-linear**.)
g. Select **OK**.
h. Select **Done**.
i. Select **Quit**.
j. Select **OK**.
k. Return to original desktop.
l. Check film and adjust camera, if necessary.

9-5 Complaint: No Picture.

- The signal cable should be completely connected to the display card/computer.
- The display card should be completely seated in its slot.
- Power switch and computer power switch should be in the ON position.
- Check the signal cable connector for bent or pushed-in pins.

9-6 Complaint: Image Persistence.

Image persistence is when a ghost of an image remains on the screen even after the monitor has been turned off. Unlike CRT monitors, the LCD monitor's image persistence is not permanent.

Note

It is still recommended that a screen saver be used whenever the screen is idle.

9-7 Complaint: Image is unstable, unfocused, or swimming is apparent.

- Signal cable should be completely attached to the computer.
- Check the monitor and your display card with respect to signal timings.
- (LCD's Only) Change the video mode to non-interlace and, if possible, use a 50 Hz refresh rate.
- Adjust Monitor controls to focus and adjust display. Refer to the vendor manual that came with the LCD Monitor for the operation of this function.
- (LCD's Only) If your text is garbled, change the video mode to non-interlace and, if possible, use a 50 Hz refresh rate.

9-8 Complaint: LED on monitor is not lit (no green or amber color can be seen).

- Power switch should be in the ON position and power cord should be connected.
- Make certain the computer is not in a power-saving mode.

9-9 Complaint: Display image is not sized properly.

Push the Auto adjustment button.

9-10 Complaint: Selected resolution is not displayed properly.

Push the Auto adjustment button.

9-11 Complaint: Diagnostic Image Quality has been traced to problem with LCD setup.

Operator has adjusted the Brightness and Contrast of the monitor which has affected Camera/Film imaging. Re-calibration may be necessary. Start with Section 1 of this procedure.

9-12 Complaint: Display image has a green cast to it.

- Select **TYP** in the OSM™ Information menu and press the **<** or **>** control button.
- Set the **Color Temperature** to 9300K.

9-13 Complaint: Display image ghosts or gray lines in areas of extreme black and white transitions.

The LCD Refresh Rate is too fast and must be slowed down to allow for the pixels to react properly. All LCD monitors should be set to Refresh Rate (also known as a Line Rate or Frame and Swap Rate) of 60Hz. The trade-off will be better black and white images at the sacrifice of color smoothness. Letters such as “V” and “W” will appear more jagged. See Section 4 of this procedure.

9-14 Complaint: Display image blurry or blotchy images described as poor grayscale.

LCD’s are highly dependent on gamma correction. Perform the gamma correction procedure to solve this problem. See Section 5 of this procedure.

Note

The off-the-shelf LCD’s on Signa are optimized for PC use from the factory, not for diagnostic imaging. The LCD monitors are not plug-and-play with regard to the Signa Applications Software and SGI Computer Video system. This requires that a complete monitor setup and calibration be performed at:

- Installation
- During a replacement of a failed monitor.
- Any time Software is reloaded.

9-15 Complaint: Poor grayscale, window or level, lack of definition.

LCD technology uses three color film plates and a backlight to display different shades of color and gray. They are highly dependent on gamma correction. The off-the-shelf LCD’s on Signa are optimized for PC use from the factory, not for diagnostic imaging. Attaching an LCD to the Signa system without calibrating it will cause image quality problems that can be interpreted as system problems, leading to unnecessary:

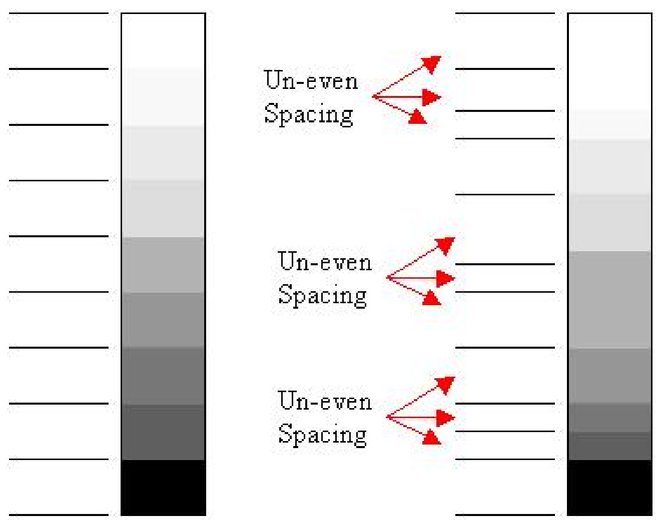
- Hardware replacement
- Protocol troubleshooting
- Customer satisfaction issues.

9-16 LCD FAQ's:

- The monitor calibration information is NOT saved during the SavInfo backup routine or accessible via Healthpage.
- There is a big difference between CRT and LCD monitors. CRT's are much less sensitive to a wrong gamma setting.
- Gamma Look-Up-Tables (LUT's) are specific to the LCD model and should only be used on the model for which they were designed.
- Gamma Look-Up-Tables (LUT's) consist of thousands of luminescent data points that are used by the electronic circuitry to generate uniform transitions from one color or grayscale level to the next.
- All recent LCD's used in Signa have an optimal single gamma value, as well as a complete Gamma Look-Up-Table to maximize flexibility when setting up the monitor.
- Always use the Gamma Look-Up-Table as a default.

9-17 What is Gamma?

Gamma normalizes the grayscale so each level is uniform in size and shape across the entire display:



GRAYSCALE COMPARISON
ILLUSTRATION 9-1

Gamma correction adjusts the grayscale or color gradient so each step is exactly the same size and shade of color or gray. If the steps are not uniform in size and color, the monitor will display the error as poor transitions from one step to the other. This appears to the eye as small ghosts or ragged edges.

9-18 How do I set the correct gamma?

When the monitor type is selected in the Install GUI, the correct gamma table (LUT) for that monitor is automatically loaded. The gamma LUT remains active even if a reboot occurs.

9-19 Can I use the gamma command to set gamma?

NO. The gamma command is no longer an effective way to permanently change gamma. This command should be used only if the need arises to temporarily change to a single gamma value, overriding any previously installed gamma LUT.

Using the gamma command on systems with an LCD will most likely CAUSE IMAGE QUALITY PROBLEMS.

Using a gamma LUT will, in most cases, produce better display image quality than a single gamma value. A single gamma value is much less effective on LCD's at providing a uniform grayscale.

9-20 How do I determine if the correct gamma table or setting is installed?

Open up the Install GUI. Look at the Display Type field. To the right of the Option box will display either a monitor type or a single number.

- If a monitor type is in this box then a Gamma Look-Up-Table for that monitor is installed.
- If a single value is displayed, then the CRT/Custom Option was used to set a single gamma value.

9-21 When I type `gamma` in a C-shell, I see a number. Is that the gamma setting?

No, it is number left over from earlier software versions and has nothing to do with the gamma setting.

9-22 When I type `gamma 1.05` in a C-shell, the LCD changes in appearance. Why?

The IRIX Operating System commands (such as `gamma`) take priority over the Signa Applications Software. The Install GUI Gamma Tools are written in IRIX script and have a lower priority. Since the IRIX Operating System gamma command was invoked with a value, that value now overrides the Applications Software. The single value of 1.05 temporarily overrides the gamma LUT until a reboot occurs. Upon reboot, the gamma LUT resumes control.

9-23 A gamma value of 1.05 is what we always used in the past. Can we still use it?

The gamma value of 1.05 is no longer used. All LCD and CRT monitors now use 0.95 which was determined to be the best default single value.

Using the gamma command on systems with an LCD will most likely CAUSE IMAGE QUALITY PROBLEMS.

Using a gamma LUT will, in most cases, produce better display image quality than a single gamma value. A single gamma value is much less effective on LCD's at providing a uniform grayscale.

9-24 My customer does not like the gamma setting that the Gamma LUT provides.

- By all means change it! Use the Install GUI. Under Display Type, select **CRT/Custom** and type in a single gamma value that you and the customer agree is satisfactory. A reboot will make the single setting permanent.

- There is a single optimized nominal gamma value for each monitor type, as well as a gamma table. Consult the troubleshooting sections within the proper monitor calibration procedures to find the nominal value for the monitor type.

Note

Using a single value affects the camera to a much higher degree than a gamma LUT does, so be prepared to adjust the camera as well.

9-25 Single value optimization technique.

To minimize the time needed to find a single gamma value that is acceptable to your customer, you can still temporarily use the IRIX O/S gamma command in a C-shell to tweak a single gamma value.

1. Type **`gamma n.nn`** (where *n.nn* is the nominal single gamma value for the monitor type). See Section 9-22. It will generate an immediate change to the monitor. Increasing the value will increase brightness and contrast together. Decreasing it will have the opposite effect.
2. Continue to adjust the gamma setting, checking images and films, until you arrive at a setting you like. Note the final value.

To make the value permanent.

3. Open the Install GUI. Under Display Type, select **CRT/Custom** and type in the new single gamma value.
4. Reboot the system to make the setting permanent.

Note:

If you open a C-shell and type `gamma`, it may not match what you installed in the Installation GUI. This is normal. See Section 9-22.

A – APPENDIX A

A-1 Focus, Screen Position, Clock Rate and Phase Adjustments

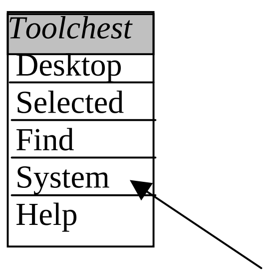
The Auto Adjust procedure done in Section 5 takes care of these for you. All other values in the Screen Manager should stay at the default values except Contrast and Display.

Use this appendix only if you see focus problems, problems with screen position relative to monitor mask, phase problems (horizontal bars sweeping through the LCD), or vertical problems.

A-2 Adjusting Focus Using LX Calibration Tools from Toolchest (Alternate)

If you cannot adjust the monitor satisfactorily with the front panel buttons and OnScreenManager, you may want to try the adjustments described in this section.

1. From the Tools Menu on Signa Open a **C-shell**.
5. Switch to root owner. (Type: **`su`** <**`Enter`**> password: **`operator`** <**`Enter`**>).
6. Type: **`[toolchest&].`** A pulldown menu will appear in the upper left-hand portion on the display.



TOOLCHEST MENU
ILLUSTRATION A-1

7. Select **System** with the right mouse button.
8. Select **Confidence Tests** from the extended menu. A Confidence Test pop-pup will appear after a few seconds.
9. From the Confidence Test pop-up, select the **Monitor Icon**. A monitor calibration menu will appear on the screen.
10. Select **Focus** from the Test Options menu by using the tab key.
11. Put the mouse pointer on the Test Options menu. Click the middle mouse button to toggle (hide) the Test Options menu.
12. Push the auto-adjustment button on the front of the monitor. This should setup Focus, Horizontal, Vertical Centering, sync to the proper Clock Rate and monitor Phase.
13. Observe the test pattern displayed. It should now be centered and in focus. This should be all that is necessary to adjust the display controller output to match the LCD monitor.
14. Toggle the Monitor Calibration toolbar back on by selecting the middle mouse button. Select the grayscale menu option.

15. Toggle the menu back off by selecting the middle mouse button again.

A-3 Setting up LCD Display for Contrast/Brightness. Using LX Calibration Tools from Toolchest (Alternate)

This alternative procedure uses the Monitor Calibration tools and patterns. Steps 1 through 4 should be done while observing the image for crisp lines, overall uniformity of pattern, and to setup the LCD monitor so it is at its brightest level without blooming the image or adding distortions. The intent of this section is to achieve the best image appearance you can with the LCD. At the completion of this section you should check the display against industry-standard SMPTE pattern. See Section 7 – LCD Calibration for Optimum Image Viewing.

1. Select the Contrast/Brightness button and press [Enter].
2. Adjust brightness to maximum level 100%.
3. Look at the monitor and adjust the Contrast and Brightness to optimize the test clarity of the test pattern. This is an iterative process between both settings.
4. When the display contrast and brightness display properly, press [Enter] to save the adjusted settings.
5. Exit the Monitor Calibration by selecting **Quit**.
6. Select **File** on the Confidence Test Window and choose **Exit**.
7. Click on the word **Toolchest** at the top of the menu with the right mouse button and select **Close**.
8. Exit the C-shell by typing: `Exit [Enter]`

A-4 Monitor Adjustments

The monitor should be positioned no closer than 16 inches and no further away than 28 inches from your eyes. The optimal distance is 24 inches for either of the monitors.

Allow the monitor to warm-up for 20 minutes before performing any adjustments.

The response time for the LCD monitor is 100 ms. It is normal to see a trail on the screen if the mouse is moved quickly.

A-5 Contrast and Brightness Adjustment for NEC1990SX LCD Monitor

1. You will need to use the SMPTE pattern for this adjustment. Refer to that section in this procedure.
2. From the Contrast and Brightness icon  use the **Control** buttons to adjust the contrast lower if the maximum setting causes visible tearing or smearing of the pattern or alphanumeric characters (items 1, 2, 5-8 in Illustration 7-3).
3. From the Contrast and Brightness icon  use the **Control** buttons to adjust the brightness of the display monitor if the 5% and 95% patches are not visible (item 6 in Illustration 7-3).

Note

It may be necessary to re-adjust the contrast if tearing or smearing of the pattern or alphanumeric characters occurs (items 1, 2, 5-8 in Illustration 7-3).

A-6 On-Screen Controls

This section describes the adjustments available for the OSM controls. The controls labeled with "•" are required during the set-up of the monitor.

- **Brightness and Contrast →** 

Brightness: Adjusts the overall image and background screen brightness.

Contrast: Adjusts the image brightness in relation to the background.

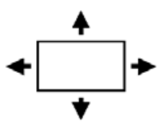
Auto Adjust Contrast: (Analog Input Only) Adjusts the image displayed for non-standard video inputs.

Auto AUTO BRIGHTNESS (Analog input only)

BRT

- **Auto Adjust →** 

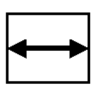
Automatically adjust the Image Position or the H. Size setting.

Position Controls → 

Position: Controls Horizontal Image Position within the display area of the LCD.

V. Position: Controls Vertical Image Position within the display area of the LCD.

Auto: Automatically sets the horizontal and vertical image position within the display area of the LCD.

- **Image Adjust Controls →** 

Size: Adjusts the horizontal size by increasing or decreasing this setting.

Fine: Improves focus, clarity, and image stability by increasing or decreasing the Fine setting.

Auto: This feature automatically adjusts the fine setting for changing signal conditions.

FINE: (Analog input only).

- **AccuColor® Control System →** 

Six color presets select the desired color setting. This setting must be set to selection 1 - **9300**. If a setting is adjusted, the name of the setting will change to Custom. NATIVE preset cannot be changed. It is the Default setting for the LCD.

R, B, G: Increases or decreases the color depending upon which is selected. The change in color will appear on screen and the increase or decrease will be shown by the color bars.

Tools 1 → 

You can choose where the OSM control image appears on the screen. Selecting OSM Location allows you to manually adjust the position**.** Normally**,** no adjustment is necessary. Use default values.

a-a	**Sharpness**	Adjustment to soften the image.
	Smoothing	(Normal) Default
	Expansion Mode	(Full Screen) Default
	Video Detect	(First Detect) Default
	DVI Selection	(Default) No effect on S-VGA systems
	OFF-TIMER	Monitor turns off after predetermined setting.

Tools 2 → 

You can choose where the OSM control image appears on the screen. Selecting OSM Location allows you to manually adjust the position**.** Normally**,** no adjustment is necessary. Use default values.

Language	English
OSM Position	Default location
OSM Turn OFF	Default
OSM Lock Out	This control completely locks out access to all OSM control functions. When attempting to activate controls while in Lock Out mode, a message screen will appear indicating that the controls are locked out. Lockout is NOT recommended.
OSM Rotation	Rotate OSM from landscape to Portrait.
Resolution Notifier	Optimal Resolution is 1280 x 1024. If ON is selected a message will appear on the screen 30 seconds, notifying the user that the resolution is NOT at 1280 x 1024.
Xyi - Resolution Notifier	This optimal resolution is 1280 x 1024. If ON is selected a message will appear on the screen 30 seconds, notifying you that the resolution is NOT at 1280 x 1024.
Hot Key	You can adjust the Contrast and brightness directly.
Factory Preset	This allows you to reset all OSM control settings back to the factory settings. Highlighting the setting and then pressing the RESET button can reset individual settings.

Information → 

Indicates the current display resolution, frequency setting, and type of Sync signal of the monitor. All items under this menu should not be changed. Use default values.

Note

Mode Change should only be used if the monitor does not recognize a resolution. This shouldn't ever be the case with our system. You can change to the appropriate resolution by selecting the Mode information and selecting (increase or decrease) the corresponding option.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Oct. 6, 2008	P. Burback	Initial Version.